

Multi-modal Bipartite Graph Structure Learning with Information Bottleneck for Micro-video Recommendation

Ying He
Tianjin University of Technology
Tianjin, China
heyingscs@email.tjut.edu.cn

Desheng Cai*
Tianjin University of Technology
Tianjin, China
caidsml@gmail.com

Shengsheng Qian
Institute of Automation, Chinese
Academy of Sciences
Beijing, China
shengsheng.qian@nlpr.ia.ac.cn

Quan Fang
School of Artificial Intelligence,
Beijing University of Posts and
Telecommunications
Beijing, China
qfang@bupt.edu.cn

Yinwei Wei
Shandong University
Jinan, China
weiyinwei@hotmail.com

Changsheng Xu
Institute of Automation, Chinese
Academy of Sciences
Beijing, China
csxu@nlpr.ia.ac.cn

Abstract

Graph-based recommender systems have become prevalent in micro-video recommendation by modeling user-item interactions as a bipartite graph. However, these methods face two inherent limitations: (1) their reliance on a fixed, pre-defined graph structure makes them susceptible to noisy interactions, and (2) the multi-modal representations they learn often contain redundant information that is not discriminative enough for the recommendation task. To overcome these issues, we propose a novel Multi-modal Bipartite Graph Structure Learning network (MBGSL), which leverages the information bottleneck principle for robust micro-video recommendation. Specifically, MBGSL first learns adaptive graph structures from multi-modal content (e.g., visual, acoustic, textual) through dedicated graph learners to mitigate noise. Then, it applies an intra-modality information bottleneck to learn minimal sufficient representations within each modality and an inter-modality information bottleneck to capture distinctive information across modalities, thereby eliminating redundancy. Furthermore, the model incorporates collaborative signals through a contrastive learning objective to guide the graph structure learning process. Extensive experiments on three real-world datasets demonstrate that MBGSL achieves state-of-the-art performance, significantly surpassing existing baselines.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Multi-modal Contents; Graph-based Micro-video Recommendation; Graph Structure Learning; Information Bottleneck

*Corresponding author



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1 Introduction

Personalized micro-video recommendation has become a crucial tool in many online micro-video sharing platforms because of the explosive growth of micro-videos. Recently, **Graph neural networks (GNNs)**, such as GCN [19], GraphSAGE [11], and HetGNN [45], have been widely studied for personalized recommendation. Benefiting from the message-passing scheme over user-item interaction graphs, follow-on methods like PinSAGE [43] and LightGCN [12] have shown promising performance in graph-based collaborative filtering. Subsequent efforts, such as MMGCN [36], MICRO [49], and LGMRec [10], have further integrated multi-modal features to enrich representations. Despite the promising performance of these GNN-based methods, they still face the following limitations.

Sensitivity to Noisy Interactions. Existing GNN-based recommenders typically rely on a fixed, predefined user-item graph for representation learning. This assumption of structural reliability is often violated in practice, as real-world bipartite graphs are inherently noisy (e.g., users may misclick uninterested micro-videos) and incomplete (e.g., some interested micro-videos have not been recommended to users). Consequently, aggregating information directly over this raw graph structure can propagate and amplify noise, ultimately degrading recommendation performance. The key insight is that user preferences are more consistently reflected in the multi-modal content of the items they interact with. As illustrated in Figure 1(a), users may interact with items due to affinities in specific modalities (e.g., for their visual appeal), and an individual user's preference may even vary across modalities for the same item. For example, user u_1 interacts with micro-videos v_1 and v_2 due to an attraction to their romantic visual content. In contrast, user u_3 interacts with v_2 and v_3 because of an interest in their tragic textual descriptions. A standard graph-based model, relying on the

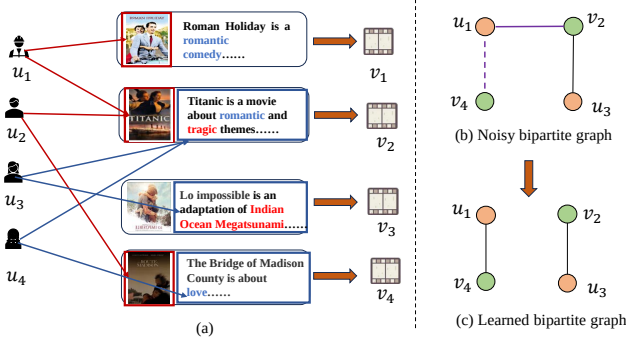


Figure 1: (a) shows the interaction data between users and micro-videos. The red arrow represents the user's interest in the visual modality and the blue arrow represents the user's interest in the textual modality of micro-videos. (b) is a noisy bipartite graph where the purple dashed line represents an incomplete edge and the purple solid line is a noisy edge. (c) is the real bipartite graph based on modality-aware user and micro-video similarity.

observed interaction graph, might detect the shared click on v_2 between u_1 and u_3 and consequently recommend v_3 to u_1 . However, this recommendation is mistaken since the underlying multi-modal content reveals that u_1 's true preference aligns more closely with consistent romantic themes instead of tragic themes, suggesting that a recommendation of v_4 would be more appropriate. Therefore, a fundamental limitation of existing methods is their inability to leverage multi-modal semantics to dynamically learn clean, task-specific bipartite graph structures that truly reflect user preferences. (as conceptualized in Figures 1(b) and 1(c)).

Redundancy in Multi-modal Representations. Addressing the first limitation also hinges on learning high-quality, compact representations. While integrating multi-modal content can enhance representations, existing methods often indiscriminately inject all available information, leading to redundant representations that diminish discriminative power. For example, GRCN [35] utilizes multi-modal contents to construct modal-specific bipartite graphs and refines the structure of each bipartite graph by the attention mechanism. However, the generated multi-modal embeddings are possibly redundant. Although recent efforts like MGGL [23] use cleaner signals to guide learning and partially filter out task-irrelevant information, their relatively simple contrastive learning frameworks struggle to yield sufficiently discriminative representations for accurate recommendation. Ideally, representations should be maximally informative for the recommendation task while being minimally redundant—an objective aligned with the information bottleneck (IB) principle. The IB principle aims to learn representations that compress input data by preserving only minimal yet sufficient information for predicting the target. Although applied in other domains, the exploration of IB for graph-based recommendation remains limited. Thus, the second limitation of current approaches is their lack of a principled mechanism to effectively learn discriminative and non-redundant representations for micro-video recommendation.

To address these limitations, we propose a novel Multi-modal Bipartite Graph Structure Learning network (MBGSL), which integrates the information bottleneck principle for robust micro-video recommendation. The framework is designed as follows: To tackle the noise sensitivity (**Limitation one**), we first construct distinct modal views and employ dedicated graph structure learners. These learners dynamically infer clean, modality-aware bipartite graph structures by measuring the representation similarity between users and micro-videos within each modality, effectively filtering out noisy interactions. To resolve the representation redundancy (**Limitation two**), we introduce a dual information bottleneck constraint. The intra-modality IB constraint compresses the information flow within each modality by minimizing the mutual information between the original noisy graph and the newly learned graph, while simultaneously maximizing the mutual information between the fused multi-modal representations and the recommendation target. This ensures the learned representations are minimal yet sufficient for the task. Complementarily, the inter-modality IB constraint minimizes the mutual information between different modal views, forcing the model to capture unique, non-redundant information from each modality and enhancing the distinctiveness of the final representations. Furthermore, we guide the structure learning with the original collaborative signals by employing a contrastive learning objective. This regularization prevents the model from generating over-sanitized graphs that diverge from real user interactions, ensuring the denoised graphs are both clean and behaviorally realistic.

The main contributions of this paper are summarized as follows:

- We propose a novel Multi-modal Bipartite Graph Structure Learning (MBGSL) network for micro-video recommendation. Our model dynamically learns clean, modality-specific bipartite graph structures to capture genuine user preferences, effectively mitigating the impact of noisy interactions.
- To learn compact and highly discriminative representations, we introduce a novel dual-constrained information bottleneck framework. This includes an intra-modality constraint to compress redundant information within each modality, and an inter-modality constraint to capture distinctive information across modalities.
- We conduct extensive experiments on three real-world datasets. The experimental results demonstrate that the proposed method achieves significant improvements on 15 state-of-the-art methods.

2 Related Work

2.1 GNN Based Recommendation

Graph neural networks (GNNs) have achieved promising performance in recommendation systems due to their powerful capability of representation learning. Early approaches like NGCF [31] and PinSAGE [30] demonstrate the power of leveraging high-order neighbor information through GNNs. Subsequent efforts such as LightGCN [12] and UltraGCN [25] further improve the efficiency and scalability of GNN by simplifying the propagation mechanism. To learn more comprehensive representations, recent research integrates multi-modal features into collaborative filtering

signals [3, 4, 10, 15, 20, 36, 39]. For example, MMGCN [36] constructs separate graphs for each modality to learn user preferences. LGMRec [10] further models local and global user dependencies for comprehensive representation learning. Although these methods have made progress, they often overlook the inherent noise in multi-modal features. To mitigate the noise issue, several recent works have introduced self-supervised learning (SSL) techniques, such as HGCL [2], MPGCL [14], and MGCL [23].

2.2 Graph Structure Learning

Although GNN-based recommenders excel at modeling user-item interaction graphs, their performance is highly sensitive to noisy interactions within the original graph structure. To address this, Graph Structure Learning (GSL) [21, 33, 48, 50] has emerged as a promising direction, aiming to learn a refined, optimal graph topology. Existing GSL methods can be broadly categorized into two lines: those that reweight edges [35] and those that jointly learn the graph structure and GNN parameters [5, 7, 16, 34, 40, 41].

However, a key limitation of these general GSL methods is their neglect of multi-modal features during the structure learning process. Some recent efforts, such as MICRO [49] and CSGCL [13], incorporate multi-modal content to construct item-item graphs. Nevertheless, they often fail to adequately filter out the noise inherent in the multi-modal features themselves. In contrast, our work introduces the Information Bottleneck principle into the multi-modal graph structure learning process. This approach is distinct in its ability to not only denoise the interaction graph but also explicitly compress intra-modal and inter-modal redundancies, thereby learning more robust and distinctive graph structures under the guidance of original collaborative signals.

2.3 Information Bottleneck

The information bottleneck (IB) principle is an approach based on information theory, which aims to learn robust representations by discarding task-irrelevant information and preserving more meaningful information of input data for downstream tasks. Specifically, given the input data X and its label Y , IB provides a framework for learning minimal sufficient representation Z . Formally, the loss function of standard IB is as follows:

$$\mathcal{L}_{IB} = -I(Z, Y) + \beta I(X, Z), \quad (1)$$

Where $I(\cdot, \cdot)$ refers to the mutual information between two random variables. β is a trade-off coefficient of the two parts.

Following IB, Alemi et al. [1] firstly introduce IB into the deep learning framework and propose variational IB to make the mutual information easy to calculate. Due to the effectiveness of IB and VIB, they are widely used in computer vision [24], natural language processing [22, 46], and reinforcement learning [8]. Besides, some works pay attention to utilizing IB and VIB for graph-based data [28, 44]. Recent studies have also explored applying it to recommendation for denoising purposes, such as GBSR [42], FairIB [38], DVIB [51], and HGIB [47]. However, our work is fundamentally different in both problem focus and technical approach. Our MBGSL framework specifically targets the noise inherent in multi-modal aware user-item interaction graphs and the redundancy within and across multi-modal representations.

3 Notations

We consider a multimodal micro-video recommendation scenario. Let $\mathcal{G}(U, V, \mathcal{E})$ denote the user-micro-video interaction bipartite graph, where $U = \{u_1, u_2, \dots, u_{|U|}\}$ is the set of user nodes, $V = \{v_1, v_2, \dots, v_{|V|}\}$ is the set of micro-video nodes, and the set $\mathcal{E} \subseteq U \times V$ represents the observed interactions between users and micro-videos, $|U|$ and $|V|$ are the numbers of users and micro-videos, respectively. This graph structure can be equivalently represented by an adjacency matrix $\mathbf{A} \in \mathbb{R}^{|U| \times |V|}$, where $a_{ij} = 1$ represents that user i has interacted with micro-video j . Each $v_j \in V$ is associated with multi-modal features $\mathbf{x}_j = \{\mathbf{x}_j^m | m \in M\}$, where $M = \{visual(v), acoustic(a), textual(t)\}$ is the modality indicator, and $\mathbf{x}_j^m$ is the raw feature vector of modality m for micro-video v_j .

4 Method

Figure 2 illustrates the overall architecture of our proposed MBGSL model. In this section, we analyze its main components in detail.

4.1 Multi-modal Bipartite Graph Structure Learning

Feature Initialization: Unlike existing Graph Structure Learning (GSL) methods that fuse multi-modal features into a unified representation, which obscure modality-specific user preferences, our approach learns a distinct bipartite graph structure for each modality. This enables the capture of fine-grained, modality-aware user interests. As depicted in Figure 2, we initialize three modality-specific graphs $\{\mathcal{G}_m = (U, V, \mathbf{X}^m, \mathbf{A})\}_{m \in M}$, each sharing the original bipartite structure $\mathbf{A}$ but retaining only the feature set $\mathbf{X}^m$ of the corresponding modality. The initial adjacency matrix for each modality is set as $\mathbf{A}^m = \mathbf{A}$, which will be iteratively refined by our graph structure learners based on node representation similarity.

For micro-video nodes, we initialize their modality-specific features using established pre-trained models. For the visual modality, the input raw images are encoded by VGGNet19, which is pre-trained on ImageNet[27], the acoustic modality is pre-trained by audioAna, and the textual modality is encoded by the pre-trained BERT [6]. The resulting initialized micro-video embeddings for modality m are denoted as $\mathbf{E}_v^m \in \mathbb{R}^{|V| \times d}$. For user nodes, which lack innate content features, we initialize their embeddings by aggregating the features of their neighboring micro-videos. We employ a summation operation for this purpose: $\mathbf{e}_u^m = \sum_{v \in \mathcal{N}_u} \mathbf{e}_v^m$.

Bipartite Graph Structure Learner: As discussed earlier, real-world user-micro-video interaction graphs are inherently noisy, and the message-passing mechanism of GNNs can amplify these noises, leading to suboptimal recommendations. To mitigate this, we design a metric learning-based structure learner to refine the original graph topology. Recognizing that user preferences are expressed differently across modalities, we learn a distinct graph structure for each modality m .

We adopt an iterative process where enhanced node embeddings lead to improved graph structures, and vice versa. We first employ a GNN encoder to learn initial node representations. While our framework is compatible with various GNNs, we opt for a LightGCN [12] backbone for its efficiency and effectiveness. Given an initialized modality-aware graph $\mathcal{G}_m = (U, V, \mathbf{A}^m)$, the LightGCN

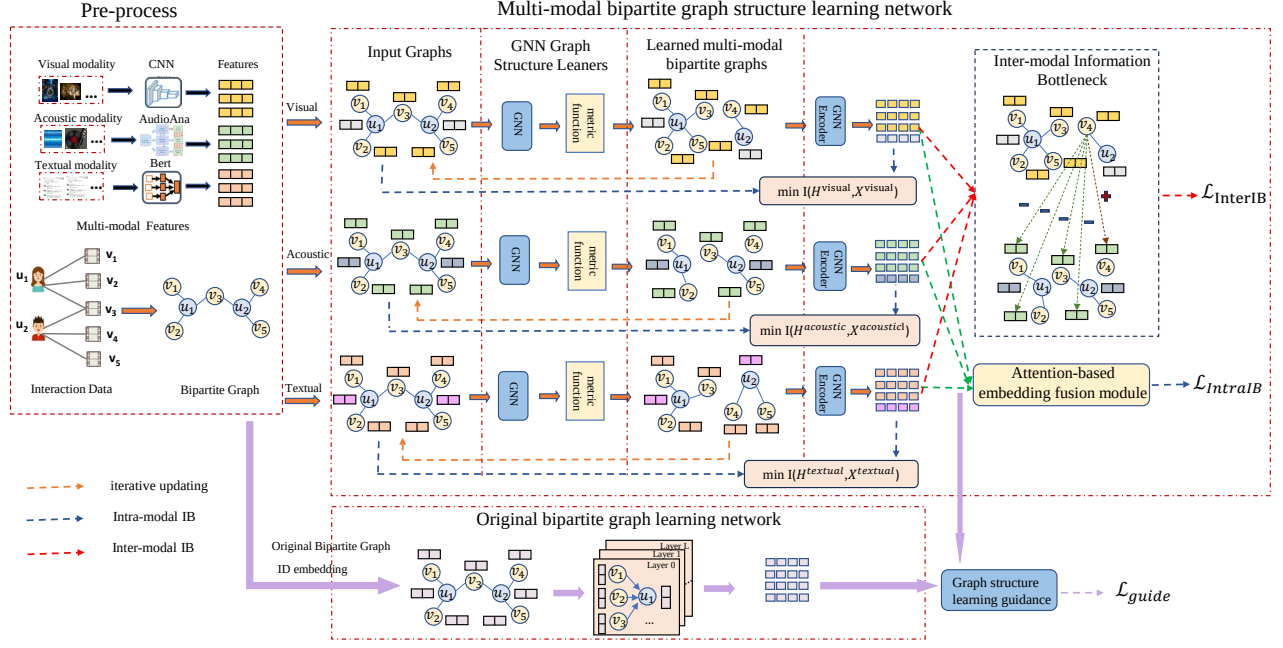


Figure 2: Illustration of our proposed MBGSL architecture.

propagation at the l -th layer is defined as:

$$\mathbf{e}_u^{m(l)} = \frac{\mathbf{e}_u^{m(l-1)}}{d_u^m + 1} + \sum_{v \in \mathcal{N}(u)} \frac{1}{\sqrt{|d_u^m + 1|} \sqrt{|d_v^m + 1|}} \mathbf{e}_v^{m(l-1)}, \quad (2)$$

$$\mathbf{e}_v^{m(l)} = \frac{\mathbf{e}_v^{m(l-1)}}{d_v^m + 1} + \sum_{u \in \mathcal{N}(v)} \frac{1}{\sqrt{|d_u^m + 1|} \sqrt{|d_v^m + 1|}} \mathbf{e}_u^{m(l-1)}, \quad (3)$$

where $\mathcal{N}(\cdot)$ denotes the neighbor set, and d_u, d_v are node degrees. $\sqrt{|d_u + 1|} \sqrt{|d_v + 1|}$ is the symmetric normalization term.

Metric function: Inspired by [5, 29], we refine the adjacency matrix A_m using a multi-head parametric metric function. This function assesses the modality-specific interests between user u and micro-video v based on their embeddings $\mathbf{e}_u^m$ and $\mathbf{e}_v^m$. The metric learning function is formulated as:

$$\mathbf{A}_{uv,k}^m = g(\mathbf{e}_u^m \odot \mathbf{w}_k \odot \mathbf{e}_v^m), \quad \mathbf{A}_{uv}^m = \frac{1}{n} \sum_{k=1}^n \mathbf{A}_{uv,k}^m, \quad (4)$$

where g is a non-linear projection head, $\odot$ is the Hadamard product, $\mathbf{w}_k$ is a learnable weight vector for the k -th head, and n is the total number of heads. A higher score in $\mathbf{A}_{uv}^m$ indicates a stronger preference of user u for micro-video v in modality m .

In the recommended scenario, the interaction matrix of users and micro-videos is usually sparse and non-negative. To meet these properties, we follow previous work [16] to modify $\mathbf{A}_{uv}^m$:

$$\tilde{\mathbf{A}}_{uv}^m = \text{NORM}(\text{SPAR}(\mathbf{A}_{uv}^m)), \quad (5)$$

where $\text{SPAR}(\cdot)$ denotes a top- k sparsification operation that retains only the k largest values per row in matrix $\mathbf{A}_{uv}^m$, setting others to 0. $\text{NORM}(\cdot) = (\tilde{\mathbf{D}}^{-\frac{1}{2}} \mathbf{A}_{uv}^m \tilde{\mathbf{D}}^{-\frac{1}{2}})$ is the symmetric normalization, which makes sure that the value of $\mathbf{A}_{uv}^m$ is in the range of $[0, 1]$, with

$\tilde{\mathbf{D}}$ being the degree matrix of $\mathbf{A}_{uv}^m$. The graph structure and node representations are then updated jointly and iteratively.

GNN-based encoder: After obtaining the refined graph structure, we employ another LightGCN encoder with separate parameters to generate the final modality-aware embeddings $\mathbf{h}_u^m, \mathbf{h}_v^m$ and the original node representations $\mathbf{h}_u^o, \mathbf{h}_v^o$.

4.2 Attention-based Embedding Fusion:

By the aggregation process of the GNN-based encoder on each modality-aware bipartite graph, we obtain the modality-aware node representations $\mathbf{h}_u^m$ and $\mathbf{h}_v^m$ for user u and micro-video v . To get the fused representation for each node, we fuse the corresponding node representations from all modalities according to their importance. Following [29], we utilize a self-attention mechanism to measure the contributions of different modalities. Take the micro-video nodes for example:

$$\mathbf{H}_v = \sum_{m \in \mathcal{M}} \alpha_m \mathbf{H}_v^m, \quad (6)$$

where α_m is the contribution value of node v in modality m to the final representation, and it can be calculated based on self-attention with residual connection:

$$\alpha_m = \text{SOFTMAX}\left(\frac{\mathbf{H}_v^m W^Q (\mathbf{H}_v^m W^K)^T}{\sqrt{d}} \mathbf{H}_v^m W^V + \mathbf{H}_v^m W^Q\right), \quad (7)$$

where $W^Q \in \mathbb{R}^{d \times d}$, $W^K \in \mathbb{R}^{d \times d}$, and $W^V \in \mathbb{R}^{d \times d}$ are different transform matrices, $\sqrt{d}$ is a scaling factor for stability. The residual term $\mathbf{H}_v^m W^Q$ helps preserve original modality-specific information during fusion. The same fusion process is applied to user representations $\mathbf{h}_u^m$ to obtain the unified user embedding $\mathbf{H}_u$.

4.3 Intra-modal Information Bottleneck

Through the above multi-modal bipartite graph structure learning process, we obtain graph structures and node representations for each modality. However, these learned representations may contain redundant information, such as the background in the visual modality. Ignoring such redundancy contained in these learned feature representations could degrade the performance of recommendation systems. To learn minimal sufficient representations, we introduce an intra-modal Information bottleneck (IB) constraint into the bipartite graph structure learning process. The IB principle aims to learn a representation H that is maximally informative about the target Y while being maximally compressive about the input X . Formally, we optimize the following objective for each modality m :

$$\max I(\mathbf{H}, \mathbf{Y}) - \lambda \sum_{m \in M} I(\mathbf{H}^m, \mathbf{X}^m), \quad (8)$$

where $I(\mathbf{H}, \mathbf{Y})$ denotes the mutual information between the target $\mathbf{Y}$ and the fused node representation $\mathbf{H}$, and $I(\mathbf{H}^m, \mathbf{X}^m)$ denotes the mutual information between the input $\mathbf{X}$ and the node representation $\mathbf{H}$ in each modality. λ is the trade-off factor.

Since directly computing mutual information is intractable, we derive the lower bound of $I(\mathbf{H}, \mathbf{Y})$ and the upper bound of $I(\mathbf{H}^m, \mathbf{X}^m)$.

lower bound of $I(\mathbf{H}, \mathbf{Y})$: Assuming that the node representation $\mathbf{H}$, the target random variable $\mathbf{Y}$, and the input random variable $\mathbf{X}$ follow the Markov Chain $\mathbf{Y} \leftrightarrow \mathbf{X} \leftrightarrow \mathbf{H}$, then we have:

$$\begin{aligned} I(\mathbf{H}, \mathbf{Y}) &\geq \int dx dy dh p(x) p(h|x) p(y|x) \log q(y|h) \\ &= \mathbb{E}_{h \sim p_\theta(h|x)} [\log q(y|h)], \end{aligned} \quad (9)$$

where $q(y|h)$ is the variational approximation distribution of $p(y|h)$, $p_\theta(h|x)$ is the output distribution of the graph neural network. We provide the detailed proof in the Appendix A.

In the micro-video recommendation scenario, the target random variable $\mathbf{Y}$ can be regarded as the random variable $\mathbf{A}^t$, where $\mathbf{A}^t$ represents the interactions between users and micro-videos. Hence, Eq. (9) can be termed as the reconstruction of the interaction $\mathbf{A}^t$:

$$I(\mathbf{H}, \mathbf{Y}) = I(\mathbf{H}_u, \mathbf{H}_v; \mathbf{A}^t) = \mathbb{E}_{p_\theta(\mathbf{h}_u|\mathbf{x}_u) p_\theta(\mathbf{h}_v|\mathbf{x}_v)} [\log p(\mathbf{a}^t|\mathbf{h}_u, \mathbf{h}_v)]. \quad (10)$$

We randomly sample N positive and negative pairs in the dataset D by Monte Carlo sampling. The mutual information $I(\mathbf{H}_u, \mathbf{H}_v; \mathbf{A}^t)$ would have an approximation lower bound formulated as the cross-entropy term:

$$I(\mathbf{H}_u, \mathbf{H}_v; \mathbf{A}^t) = \sum_{(u,v) \in \mathcal{R}} \log(z(\mathbf{h}_u, \mathbf{h}_v)) + \sum_{(u,v') \notin \mathcal{R}} \log(1 - z(\mathbf{h}_u, \mathbf{h}_{v'})), \quad (11)$$

where $z(\cdot)$ is a scoring function.

upper bound of $I(\mathbf{H}^m, \mathbf{X}^m)$: Let $r(\mathbf{h}^m)$ be a variational approximate distribution of the prior $p(\mathbf{h}^m)$, we derive the upper bound of $I(\mathbf{H}^m, \mathbf{X}^m)$:

$$I(\mathbf{H}^m, \mathbf{X}^m) \leq \int dx dh p(\mathbf{x}^m) p_\theta(\mathbf{h}^m|\mathbf{x}^m) \log \frac{p_\theta(\mathbf{h}^m|\mathbf{x}^m)}{r(\mathbf{h}^m)}. \quad (12)$$

We also provide the detailed proof in the Appendix A.

By using the Monte Carlo sampling method, the upper bound can be estimated as:

$$\begin{aligned} I(\mathbf{H}^m, \mathbf{X}^m) &\approx \frac{1}{N} \sum_{i=1}^N \int dh p_\theta(\mathbf{h}^m|\mathbf{x}_i^m) \log \frac{p_\theta(\mathbf{h}^m|\mathbf{x}_i^m)}{r(\mathbf{h}^m)} \\ &= \mathbb{E}_{\mathbf{x}^m} [D_{\text{KL}}(p_\theta(\mathbf{h}^m|\mathbf{x}^m) || r(\mathbf{h}^m))]. \end{aligned} \quad (13)$$

To compute the analytic KL-divergence above, we assume that the prior distribution $r(\mathbf{h}^m)$ is a standard Gaussian distribution $\mathcal{N}(0, I)$, and the posterior $p_\theta(\mathbf{h}^m|\mathbf{x}_i^m)$ is a Gaussian distribution whose mean μ and variance Σ are parameterized by our modality-aware GNN-based encoder $f_\theta(\mathbf{x}^m)$ followed by MLPs:

$$\mu^m = \text{MLP}_{\phi_\mu}(f_\theta(\mathbf{x}^m)), \quad \Sigma^m = \text{MLP}_{\phi_\Sigma}(f_\theta(\mathbf{x}^m)). \quad (14)$$

In order to realize the back-propagation for stochastic optimization of the sampling operation from Gaussian distribution, we utilize the reparameterization trick [18] to calculate h :

$$\mathbf{h}^m = \mu^m + \Sigma^m \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad (15)$$

where $\odot$ is the element-wise product. ϵ is sampled from $\mathcal{N}(0, I)$.

The KL divergence can then be computed in closed form. For a sample of N nodes:

$$I(\mathbf{H}^m, \mathbf{X}^m) \approx \frac{1}{N} \sum_{i=1}^N [D_{\text{KL}}(\mathcal{N}(\mu_i^m, \Sigma_i^m) || \mathcal{N}(0, I))]. \quad (16)$$

Combining the two bounds, the final loss for the intra-modal IB:

$$\begin{aligned} \mathcal{L}_{\text{IntraIB}} &= - \sum_{(u,v) \in \mathcal{R}} \log(z(\mathbf{h}_u, \mathbf{h}_v)) - \sum_{(u,v') \notin \mathcal{R}} \log(1 - z(\mathbf{h}_u, \mathbf{h}_{v'})) \\ &\quad + \lambda \sum_{m \in M} \frac{1}{N} \sum_{i=1}^N [D_{\text{KL}}(\mathcal{N}(\mu_i^m, \Sigma_i^m) || \mathcal{N}(0, I))], \end{aligned} \quad (17)$$

where λ controls the trade-off between predictive power and representation compression.

4.4 Inter-modal Information Bottleneck

Fused feature representations of micro-videos reflect users' preferences in different aspects. Intuitively, modality-specific node representations should preserve distinctive characteristics and filter out redundant information across modalities. Although existing methods like GRCN [35] model multi-modal features by constructing modality-specific interaction graphs, they fail to reduce such cross-modal redundancy. To address this, we propose an inter-modal information bottleneck constraint to extract distinctive characteristics for each modality by minimizing the mutual information between node representations of different modalities. The objective is defined as:

$$\arg \min_{\theta} \sum_{m, m' \in M} I(\mathbf{H}_{\theta_m}^m, \mathbf{H}_{\theta_{m'}}^{m'}), \quad (18)$$

where $\mathbf{H}^m$ denotes the node representations in modality m , and θ_m represents the parameters of the encoder for modality m .

Since the mutual information is intractable, we draw inspiration from contrastive learning but invert its objective. Specifically, we structure our loss to minimize the alignment between modalities relative to other nodes in the batch. The loss function for a batch of nodes $\mathcal{B}$ is formulated as:

$$\mathcal{L}_{\text{InterIB}} = \sum_{m, m' \in M} \frac{1}{|\mathcal{B}|} \sum_{v \in \mathcal{B}} -\log(1 - \frac{\exp(\text{sim}(\mathbf{h}_v^m, \mathbf{h}_v^{m'})/\tau_1)}{\sum_{i \in \mathcal{B}} \exp(\mathbf{h}_v^m \cdot \mathbf{h}_i^{m'}/\tau_1)}), \quad (19)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity function, and τ_1 is the temperature hyper-parameter.

4.5 Bipartite Graph Structure Learning Guidance

We employ a contrastive learning framework to align the learned modality-specific representations with those from the original graph. To enhance robustness, we first apply random augmentations to both the modality-aware graphs and the original graph. For modality-aware graphs, we randomly mask a portion of node features and obtain $\mathbf{H}^m = \mathbf{H}^m \odot \mathbf{M}_{mf}$, where $\mathbf{M}_{mf}$ is a binary mask matrix with each entry sampled from a Bernoulli distribution. For the original graph, we randomly do an edge dropout operation and get the new graph structure $\mathcal{G}' = (\mathbf{V}, \mathbf{M}_{ed} \odot \mathbf{A})$, where $\mathbf{M}_{ed}$ is generated by Bernoulli distribution with each matrix entry m_{ed} is similarly sampled from a Bernoulli distribution.

We then maximize the mutual information between corresponding nodes across the augmented modality-aware graphs and the original graph through the following contrastive loss:

$$\mathcal{L}_{guide} = \sum_{m \in M} \sum_{v \in V} -\log \frac{\exp(\text{sim}(\mathbf{h}_v^m, \mathbf{h}_v^o)/\tau_2)}{\sum_{i \in V} \exp(\text{sim}(\mathbf{h}_v^m, \mathbf{h}_i^o)/\tau_2)}, \quad (20)$$

where τ_2 is the temperature hyper-parameter.

4.6 Model Optimization

To optimize our model, we combine the graph structure learning guidance loss $\mathcal{L}_{guide}$ with the intra-modal information bottleneck loss $\mathcal{L}_{IntraIB}$ and the inter-modal information bottleneck loss $\mathcal{L}_{InterIB}$, which is formulated as the final loss $\mathcal{L}$:

$$\mathcal{L} = \gamma_1 \mathcal{L}_{guide} + \mathcal{L}_{IntraIB} + \gamma_2 \mathcal{L}_{InterIB} + \beta \|\Theta\|_2^2, \quad (21)$$

where $\|\Theta\|_2^2$ is the L_2 regularization term to avoid overfitting, γ_1 , γ_2 and β are their corresponding coefficients.

5 Experiments

To comprehensively evaluate the effectiveness of our proposed model, we conduct experiments on three real-world datasets including Kwai, Tiktok and MovieLens (MLs) and compare MBGSL with several representative baselines: Traditional method: **DeepFM** [9]; GNN-based methods: **PinSAGE** [43], **LightGCN** [12], **HAN** [32], and **HHFAN** [4]; Graph contrastive learning (GCL) based methods: **SGL** [37], **HGCL** [2], **MPGCL** [14]; Graph structure learning methods: **IDGL** [5], **GRCN** [35], **MICRO** [49], and **CSGCL** [13]; Multi-modal method: **MGCL** [23], **LGMRec** [10], and **COHESION** [39]. The details of these baselines are described in the Appendix B.

5.1 Evaluation Metrics and Implementation

In this work, for each dataset, the split ratios of training set, validation set, and test set are 6:2:2 and 4:3:3, respectively. We leverage three commonly used Top-K recommendation metrics, which are the F value measure at top K (F@K), Normalized Discounted Cumulative Gain at top K (NDCG@K), and Area Under the ROC Curve (AUC), to evaluate our proposed model and baselines. We set $K=10$ and report the average performance on the test set. We leverage the Adam[17] optimizer to optimize our proposed model. The learnable weights of our proposed model are initialized using a Gaussian

distribution with a mean of 0 and a standard deviation of 0.01. The learning rate l is chosen from the set $\{1.0 \times 10^{-1}, 3.0 \times 10^{-1}, 5.0 \times 10^{-1}, 1.0 \times 10^{-2}, 3.0 \times 10^{-2}, 5.0 \times 10^{-2}, 7.0 \times 10^{-2}, 1.0 \times 10^{-3}, 3.0 \times 10^{-3}, 5.0 \times 10^{-3}, 7.0 \times 10^{-3}, 1.0 \times 10^{-4}, 3.0 \times 10^{-4}, 5.0 \times 10^{-4}, 7.0 \times 10^{-4}, \}$ and the regularization coefficient β with L2 is chosen from the set $\{3.0 \times 10^{-2}, 5.0 \times 10^{-2}, 3.0 \times 10^{-3}, 5.0 \times 10^{-3}, 3.0 \times 10^{-4}, 5.0 \times 10^{-4}, 3.0 \times 10^{-5}, 5.0 \times 10^{-5}\}$. The hyper-parameters $\gamma_1 = 0.5$, $\gamma_2 = 0.7$, $\lambda = 0.9$, $n = 6$ and $\beta = 0.005$. All models, including baselines and our MBGSL, use a fixed embedding size of 200 for fair comparison.

5.2 Experimental Analysis

The results for the 6:2:2 and 4:3:3 training ratios are shown in Table 1 and Appendix Table 3, respectively. From these results, we have **the following key observations**:

Compared with the traditional deep learning based method (i.e., DeepFM), GNN-based methods achieve good performance on three datasets, indicating that capturing complex high-order interactions between users and micro-videos is important for learning high-quality user and micro-video embeddings for recommendations. Contrastive learning based recommendation methods significantly outperform the traditional supervised GNN-based methods, showing the importance of introducing contrastive learning into GNN. For example, MPGCL improves the robustness of node representation learning by learning an encoder that is invariant to edge perturbations.

Graph structure learning based methods outperform GNN-based methods with significant performance improvement, validating the importance of adaptively refining graph topology to mitigate noise. Particularly, MICRO and GRCN, which incorporate multi-modal signals, outperform IDGL, highlighting the usefulness of content information in guiding structure learning.

Multi-modal methods, which integrate multi-modal features into graph learning, achieves highly competitive results. Their performances verify the usefulness of aligning multi-modal semantics with collaborative signals. However, their inability to explicitly denoise the graph structure limits further gains.

Our MBGSL consistently achieves the best performance across all datasets. The improvements can be attributed to its integrated design: Compared to GNN-based methods, MBGSL goes beyond message passing on a fixed graph; instead, it dynamically learns clean, modality-aware graph structures. Compared to contrastive learning based methods, it not only enhances robustness but also reconstructs the graph itself in each modality. Compared to graph structure learning methods, our approach not only learns distinct, denoised graph structures for each modality to precisely capture modality-specific user preferences, but also incorporates a dual information bottleneck to compress redundant information, achieving a more comprehensive refinement. Compared to multi-modal methods, it explicitly unifies graph denoising and redundancy reduction, leading to more discriminative and robust representations.

5.3 Ablation Studies

To investigate the contribution of each component of our MBGSL model, we conduct several ablation studies. The results are demonstrated in Table 2 and Appendix Table 4.

Table 1: Evaluation experimental results on three datasets in terms of three metrics. (K=10, training_rate=0.6)

Method	Kwai			Tiktok			MLs		
	F	NDCG	AUC	F	NDCG	AUC	F	NDCG	AUC
DeepFM	0.1827	0.3425	0.6448	0.2113	0.3376	0.6636	0.2027	0.3771	0.6229
PinSage	0.2383	0.3899	0.7037	0.2668	0.3779	0.7071	0.2749	0.4003	0.6441
LightGCN	0.3093	0.4099	0.7371	0.3019	0.4039	0.7110	0.3084	0.4117	0.6991
HAN	0.3728	0.4304	0.7441	0.3459	0.4229	0.7309	0.3541	0.4339	0.7200
HHFAN	0.4003	0.4558	0.7658	0.3836	0.4665	0.7613	0.4001	0.4682	0.7499
SGL	0.3553	0.4299	0.7231	0.3198	0.4119	0.7098	0.3538	0.4358	0.7114
HGCL	0.4100	0.4559	0.7704	0.3991	0.4799	0.7747	0.4108	0.4772	0.7601
MPGCL	0.4298	0.4711	0.7803	0.4125	0.4874	0.7871	0.4289	0.4851	0.7863
IDGL	0.3154	0.4193	0.7400	0.3291	0.4002	0.7119	0.3358	0.4201	0.7025
GRCN	0.3609	0.4301	0.7357	0.3229	0.4011	0.7198	0.3309	0.4289	0.7135
MICRO	0.3885	0.4295	0.7536	0.3598	0.4670	0.7438	0.3899	0.4638	0.7502
CSGCL	0.4314	0.4658	0.7794	0.4282	0.4862	0.7907	0.4302	0.4839	0.7874
MGCL	0.4281	0.4558	0.7695	0.4194	0.4893	0.7810	0.4229	0.4793	0.7787
LGMRec	0.4305	0.4625	0.8109	0.4257	0.4803	0.7871	0.4362	0.4772	0.7895
COHESION	0.4336	0.4701	0.7985	0.4200	0.4899	0.7905	0.4276	0.4863	0.7886
MBGSL	0.4451	0.4804	0.7937	0.4440	0.5003	0.8043	0.4553	0.5009	0.8002

Table 2: Performance comparisons between model variants on three datasets. (K=10, training_rate=0.6))

Dataset	Kwai			Tiktok			MLs		
	F	NDCG	AUC	F	NDCG	AUC	F	NDCG	AUC
MBGSL _{gl}	0.4194	0.4583	0.7784	0.4197	0.4803	0.7847	0.4312	0.4828	0.7784
MBGSL _{fm}	0.4226	0.4438	0.7552	0.4116	0.4697	0.7815	0.4303	0.4596	0.7759
MBGSL _{ig}	0.4003	0.4417	0.7531	0.4099	0.4685	0.7694	0.4017	0.4591	0.7681
MBGSL _{ir}	0.4283	0.4672	0.7681	0.4225	0.4711	0.7748	0.4084	0.4728	0.7704
MBGSL _{ia}	0.4237	0.4592	0.7614	0.4103	0.4759	0.7718	0.4195	0.4681	0.7559
MBGSL	0.4451	0.4804	0.7937	0.4440	0.5003	0.8043	0.4553	0.5009	0.8002

5.3.1 Ablation Studies of the Graph Structure Learning. We first study the influence of our proposed modality-aware graph structure learning module, including graph structure learner, multi-modal features, and the graph structure learning guidance module. In specific, we perform the following variants. **MBGSL_{gl}** removes the GNN-based graph structure learner, using only the original user-micro-video bipartite graph as input for each modality. **MBGSL_{fm}** ignores modality-aware user preferences by directly fusing multi-modal features to learn a unified bipartite graph structure. **MBGSL_{ig}** disables the graph structure learning guidance by removing the contrastive alignment between node embeddings from the original graph and those from the learned graph. Results of MBGSL and its variants across all datasets are summarized in Table 2. From the results, we can find that:

Impact of the graph structure learner component: MBGSL_{gl} consistently underperforms MBGSL across all settings, confirming that refining the original graph structure effectively mitigates the adverse effect of noisy edges on GNN-based representation learning, thereby enhancing recommendation accuracy.

Impact of the multi-modal features: The significant performance drop in MBGSL_{fm} reveals the limitation of directly fusing

multi-modal features. Modeling user preferences in a modality-aware manner is essential for capturing fine-grained user intents and improving recommendation performance.

Impact of the graph structure learning guidance component: MBGSL_{ig} is dramatically inferior to MBGSL, validating the importance of guiding the structure learning process with original collaborative signals. Without such guidance, the model risks generating over-sanitized graph structures that deviate from real user behavior patterns.

5.3.2 Ablation Studies of Information Bottleneck Objectives. We further analyze the contribution of the inter-modal and intra-modal information bottleneck (IB) constraints using the following ablation studies. **MBGSL_{ir}** removes the inter-modal IB constraint, which means that the redundancy information between modalities and the distinctive characteristic for each modality is ignored. **MBGSL_{ia}** disables the intra-modal IB objective, replacing it with a standard BPR loss [26], thereby failing to compress redundant information within each modality. As shown in in Table 2, we have the following observations:

Impact of the inter-modal information bottleneck: The performance of MBGSL_{ir} sharply decreases compared with MBGSL,

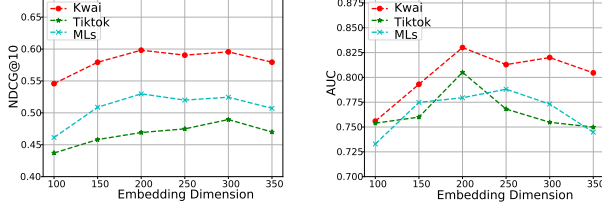
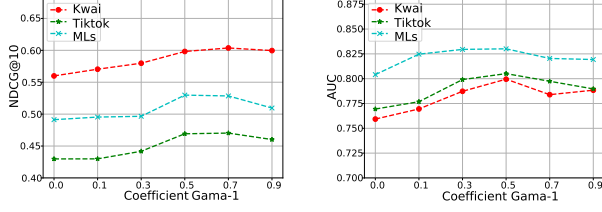


Figure 3: Performance with varying embedding dimensions

Figure 4: Performance with varying parameter Coefficient γ_1

which highlights the critical role of the inter-modal IB in encouraging modality-specific encoders to learn distinctive and non-redundant representations in each modality.

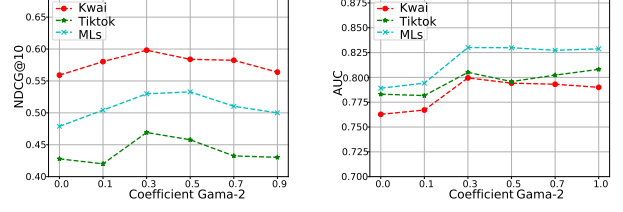
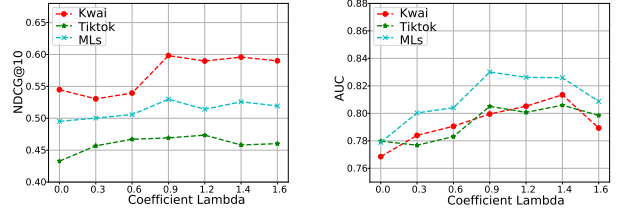
Impact of the intra-modal information bottleneck: The performance of MBGSL_{ia} greatly drops compared with MBGSL, which confirms the presence of significant redundancy in raw multi-modal features. The intra-modal IB effectively compresses this redundancy, leading to more minimal and sufficient representations that are highly beneficial for the recommendation task.

5.4 Parameter Sensitivity

We conduct a series of experiments to evaluate the sensitivity of MBGSL's key hyperparameters. After searching a wide range of values, we report the results of MBGSL on representative settings under the 80% training ratio in Figure 3-Figure 6. From the results, we find that:

(1) **Impact of the embedding dimension:** We vary the embedding dimension from 100 to 350 and present the results in Figure 3. The performance of MBGSL improves as the dimension increases from 100 to 200 and then decreases when it is larger than 200. This suggests that an appropriate dimension enhances model capacity, while an excessively large one may lead to over-smoothing or overfitting, thereby degrading recommendation quality.

(2) **Impact of γ_1 and γ_2 :** Parameters γ_1 and γ_2 control the strength of the graph structure learning guidance module and the inter-modal IB constraint, respectively. As presented in Figure 4 and Figure 5, both γ_1 and γ_2 gradually make MBGSL achieve the best results and then decline. For γ_1 , a value too small fails to adequately anchor the learned structures to the original collaborative graph, while a value too large overly constrains the denoising process, preventing effective noise removal. For γ_2 , a small value limits the model's ability to learn distinctive inter-modal features, while too large values would hurt the common features too much.

Figure 5: Performance with varying parameter Coefficient γ_2 Figure 6: Performance with varying parameter Coefficient λ

(3) **Impact of λ :** We investigate the impact of the trade-off factor λ in Eq. (17). From Figure 6, we see that performance improves as λ increases from 0 to 0.9, confirming that compressing redundant information within modalities is beneficial. However, beyond 0.9, performance degrades, likely because an overly strong compression discards valuable task-relevant information along with the redundancy.

6 Conclusion

In this paper, we propose a novel multi-modal bipartite graph structure learning network with information bottleneck for micro-video recommendation. To address the problem of noisy interactions, our model dynamically learns clean, modality-aware graph structures by leveraging similarities in multi-modal content, effectively filtering out noisy edges. Furthermore, we introduced a dual information bottleneck constraint, which includes inter-modal and intra-modal information bottleneck objectives, to explicitly reduce redundancy and learn minimal sufficient representations that are highly discriminative for the recommendation task. Extensive experimental results on three real-world datasets verify the effectiveness of MBGSL. In future work, we plan to explore more fine-grained modeling of multi-modal feature relationships and investigate the integration of external knowledge sources to further enhance recommendation performance.

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A PROOFS

Proof of Lower Bound for $I(\mathbf{H}, \mathbf{Y})$. According to the definition of mutual information, we have:

$$\begin{aligned} I(\mathbf{H}, \mathbf{Y}) &= D_{KL}(p(h, y) || p(h)p(y)) = \int dy dh p(h, y) \log \frac{p(h, y)}{p(y)p(h)} \\ &= \int dy dh p(y, h) \log \frac{p(y|h)}{p(y)}, \end{aligned} \quad (22)$$

since $p(y|h)$ is intractable, we approximate it with a variational posterior $q(y|h)$. The KL divergence between $p(y|h)$ and $q(y|h)$ is non-negative, we have:

$$\begin{aligned} I(\mathbf{H}, \mathbf{Y}) &\geq \int dy dh p(y, h) \log \frac{q(y|h)}{p(y)} \\ &= \int dy dh p(y, h) \log q(y|h) - \int dy dh p(y, h) \log p(y) \\ &= \int dy dh p(y, h) \log q(y|h) + \mathbb{H}(\mathbf{Y}) \\ &= \mathbb{E}_{p(h|x)} [\log q(y|h)] + \mathbb{H}(\mathbf{Y}). \end{aligned} \quad (23)$$

Since $\mathbb{H}(\mathbf{Y})$ is constant with respect to our model parameters, we obtain the variational lower bound:

$$I(\mathbf{H}, \mathbf{Y}) \geq \mathbb{E}_{p(h|x)} [\log q(y|h)]. \quad (24)$$

Proof of Upper Bound for $I(\mathbf{H}^m, \mathbf{X}^m)$. From the definition of mutual information, we have:

$$\begin{aligned} I(\mathbf{H}^m, \mathbf{X}^m) &= D_{KL}(p(h^m, x^m) || p(h^m)p(x^m)) \\ &= \int dh dx p(h^m, x^m) \log \frac{p(h^m, x^m)}{p(h^m)p(x^m)} \\ &= \int dx^m dh^m p(h^m, x^m) \log \frac{p(h^m|x^m)}{p(h^m)}. \end{aligned} \quad (25)$$

since the prior $p(h^m)$ is intractable, we approximate it with a variational prior $r(h^m)$. According to the fact that KL-divergence is always positive, i.e. $D_{KL}(p(h^m) || r(h^m)) \geq 0$, we have:

$$\begin{aligned} I(\mathbf{H}^m, \mathbf{X}^m) &\leq \int dx dh p(h^m, x^m) \log \frac{p(h^m|x^m)}{r(h^m)} \\ &= \mathbb{E}_{p(x^m)} \left[\int dh p(h^m|x^m) \log \frac{p(h^m|x^m)}{r(h^m)} \right] \\ &= \mathbb{E}_{p(x^m)} [D_{KL}(p(h^m|x^m) || r(h^m))], \end{aligned} \quad (26)$$

hence, we have the upper bound of $I(\mathbf{H}^m, \mathbf{X}^m)$.

B Compared Methods

The details of compared methods are as follows:

- **DeepFM**: DeepFM is a classical deep learning based recommendation method that integrates factorization machines and multi-layer perceptrons to model low-order and high-order feature interactions for representation learning.
- **PinSAGE**: PinSAGE designs an efficient random walk strategy to sample neighbors and employs a GCN to learn node embeddings, capturing both graph structure and node features..
- **LightGCN**: LightGCN is a lightweight and widely adopted GCN variant that removes nonlinear activations and feature transformations to streamline the aggregation process.

- **HAN**: HAN learns node embedding by performing the node-level attention based aggregation and the semantic-level attention based aggregation according to the pre-defined meta-paths on heterogeneous graphs.
- **HHFAN**: HHFAN develops a random walk strategy to sample different types of neighbors for a particular node and applies hierarchical feature aggregation to integrate multi-typed neighbor information.
- **SGL**: SGL is a self-supervised recommendation method, which enhances GNNs by introducing various types of graph augmentation methods, such as node/edge dropout and random walk, to performing contrastive learning.
- **HGCL**: HGCL applies graph contrastive learning networks on the constructed node-type aware homogeneous graphs. The heterogeneity and local-global information of the bipartite graph are well modeled.
- **MPGCL**: This method generates homogeneous graphs via meta-paths and applies data augmentation (e.g., node/edge perturbation) to build multiple views for contrastive learning, improving robustness to noisy edges.
- **IDGL**: IDGL is a joint learning framework that iteratively refines graph structure and node embeddings via metric-based homogeneous graph learning.
- **GRCN**: GRCN reweights user-item edges in modality-specific graphs by estimating preference confidence, and applies GCN on the refined graph to softly prune noisy interactions.
- **MICRO**: MICRO constructs modality-aware item-item semantic graphs from multi-modal content and incorporates both shared and specific modal information for comprehensive representation learning in multimedia recommendation.
- **CSGCL**: CSGCL leverages high-quality node representations via contrastive learning on both collaborative user-video interactions and semantic multi-modal features.
- **MGCL**: MGCL is a recent GCN-based multimedia recommendation model that introduces contrastive learning between modality-aware graphs and collaborative signals to reduce irrelevant noise.
- **LGMRec**: LGMRec proposes a local-global graph learning framework that combine collaborative and multi-modal signals, and captures global user interests through hypergraph embedding.
- **COHESION**: COHESION introduces a graph convolutional network with dual-stage fusion, to capture heterogeneous and homogeneous relationships.

C ADDITIONAL EXPERIMENT

(1) **Impact of the ranking position K**: We study the influence of ranking position K by varying it from 2 to 14. As shown in Figure 7, the performance of MBGSL is superior to the compared methods in most cases, which further verifies the effectiveness of MBGSL. Notably, when K is greater than 6, MBGSL achieves stable and superior performance on all datasets, indicating that its top-K recommendation lists contain a higher concentration of micro-videos aligned with user interests.

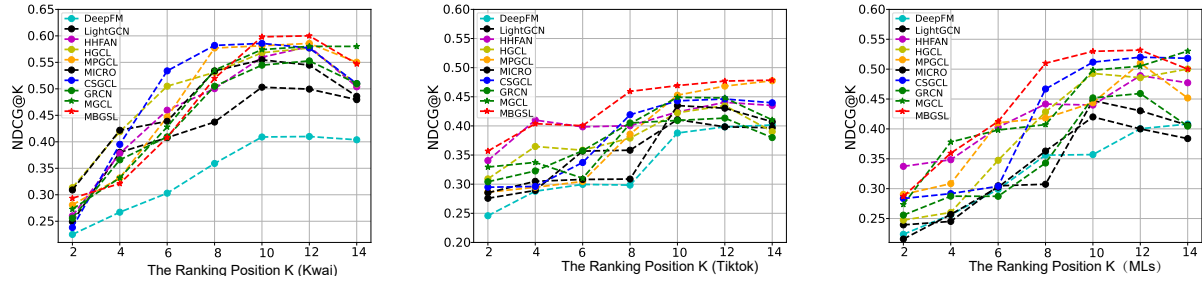
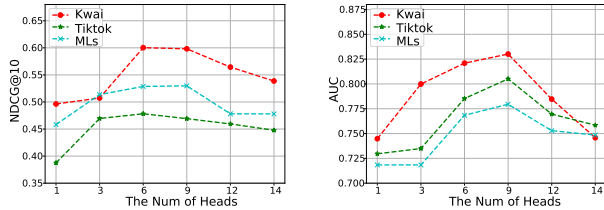
(2) **Impact of the number of head n**: We analyze the effect of the multi-head parameter n in Eq. (4) by testing values from 1 to 14. From Figure 8, we observe that the best performance is

Table 3: Evaluation experimental results on three datasets in terms of three metrics. (K=10, training_rate=0.4)

Method	Kwai			Tiktok			MLs		
	F	NDCG	AUC	F	NDCG	AUC	F	NDCG	AUC
DeepFM	0.1433	0.3073	0.6633	0.1773	0.3629	0.6104	0.2246	0.3541	0.6100
PinSage	0.2567	0.4145	0.7458	0.2599	0.4098	0.6994	0.2713	0.3981	0.6559
LightGCN	0.3988	0.4228	0.7647	0.3618	0.4224	0.7271	0.3284	0.4095	0.6871
HAN	0.3925	0.4599	0.7691	0.3894	0.4491	0.7535	0.3588	0.4413	0.7014
HHFAN	0.4389	0.4774	0.8003	0.4023	0.4525	0.7551	0.3741	0.4438	0.7399
SGL	0.3991	0.4413	0.7492	0.3792	0.4001	0.7331	0.3379	0.4300	0.7095
HGCL	0.4391	0.4803	0.8052	0.4148	0.4638	0.7599	0.3961	0.4511	0.7448
MPGCL	0.4499	0.5001	0.8102	0.4339	0.4889	0.7613	0.4227	0.4614	0.7641
IDGL	0.4008	0.4331	0.7731	0.3882	0.4331	0.7304	0.3635	0.4229	0.7093
GRCN	0.4055	0.4639	0.7748	0.3994	0.4389	0.7452	0.3701	0.4569	0.7047
MICRO	0.4391	0.4494	0.7795	0.4098	0.4692	0.7628	0.3723	0.4337	0.7558
CSGCL	0.4536	0.4999	0.8145	0.4322	0.4791	0.7713	0.4126	0.4673	0.7681
MGCL	0.4545	0.5103	0.7994	0.4284	0.4778	0.7725	0.4303	0.4665	0.7578
LGMRec	0.4492	0.5121	0.8109	0.4331	0.4768	0.7652	0.4241	0.4557	0.7791
COHESION	0.4501	0.4984	0.7985	0.4239	0.4805	0.7794	0.4326	0.4569	0.7683
MBGSL	0.4689	0.5264	0.8259	0.4505	0.5003	0.7896	0.4458	0.4799	0.7886

Table 4: Performance comparisons between model variants on three datasets. (K=10, training_rate=0.4)

Dataset	Kwai			Tiktok			MLs		
	F	NDCG	AUC	F	NDCG	AUC	F	NDCG	AUC
MBGSL _{gl}	0.4493	0.5102	0.8112	0.4301	0.4829	0.7731	0.4256	0.4668	0.7703
MBGSL _{fm}	0.4305	0.5004	0.8002	0.4093	0.4675	0.7522	0.3916	0.4461	0.7558
MBGSL _{ig}	0.4271	0.4893	0.8005	0.4071	0.4782	0.7631	0.4092	0.4558	0.7541
MBGSL _{ir}	0.4288	0.4857	0.7979	0.4285	0.4693	0.7684	0.3995	0.4373	0.7616
MBGSL _{ia}	0.4185	0.4983	0.7933	0.4168	0.4799	0.7578	0.4004	0.4484	0.7539
MBGSL	0.4689	0.5264	0.8259	0.4505	0.5003	0.7896	0.4458	0.4799	0.7886

**Figure 7: Results comparison when varying parameter K of MBGSL under NDCG on datasets****Figure 8: Performance with varying parameter n**

generally achieved at 9, revealing that a proper number of heads would improve the performance of recommendation and make the training process more stable.